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Multi-scale, multi-dimensional binocular endoscopic image depth estimation network

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ABSTRACT

During invasive surgery, the use of deep learning techniques to acquire depth information from lesion sites in real-time is hindered by the lack of endoscopic environmental datasets. This work aims to develop a high-accuracy three-dimensional (3D) simulation model for generating image datasets and acquiring depth information in real-time. Here, we proposed an end-to-end multi-scale supervisory depth estimation network (MMDENet) model for the depth estimation of pairs of binocular images. The proposed MMDENet highlights a multi-scale feature extraction module incorporating contextual information to enhance the correspondence precision of poorly exposed regions. A multi-dimensional information-guidance refinement module is also proposed to refine the initial coarse disparity map. Statistical experimentation demonstrated a 3.14% reduction in endpoint error compared to state-of-the-art methods. With a processing time of approximately 30fps, satisfying the requirements of real-time operation applications. In order to validate the performance of the trained MMDENet in actual endoscopic images, we conduct both qualitative and quantitative analysis with 93.38% high precision, which holds great promise for applications in surgical navigation.

1. Introduction

Modern medical endoscopes possess miniature cameras to check human internal conditions. It has become a standard tool for the diagnosis of diseases of the stomach, intestine, ear, nose, throat, etc. The use of endoscopes to observe these internal organs results in tiny wounds and considerably decreases the injury to the patient during medical examination or surgery due to the tiny endoscopic cameras. During endoscope-assisted surgery, a surgeon can monitor endoscopic video corresponding to the location of the lesion in real-time and then operate on the lesions. However, most contemporary endoscopes are capable of providing only two-dimensional (2D) images/videos and rely on the surgeon's experience to localize the lesion's location within the 3D space. Some immature doctors may generate abrasions or strains to the tissues during the operation and may even miss some of the polyps [1,2]. The lack of 3D information in endoscopy increases diagnostic and treatment challenges for doctors and may increase pain and risk of disease progression for patients. The addition of depth information in addition to standard 2D endoscopic images allows for the promotion of more delicate surgical procedures for complicated internal organs with higher surgical precision in a shorter operative time. In addition, 3D reconstruction of medical images utilizing depth information is useful for the classification of geometric lesions and increases the contrast of conventional images [3,4]. Thus, stereoscopic endoscopes that can satisfy the accuracy and real-time requirements of real-world medical operations have become a compelling topic for the development [5,6] of modern electronic endoscopes.

Depth estimation plays a crucial role in the development of stereoscopic endoscopy. Due to the rapid development of deep learning, the use of convolutional neural networks (CNNs) for depth estimation has led to numerous encouraging results in computer vision. Many of the models are supervised and trained on natural scene datasets containing depth map labels, e.g., several works have achieved excellent results in terms of both accuracy and speed in the autonomous driving domain with the KITTI dataset [7,8]. In addition, some of these methods are portable to the endoscopic setting. The majority of these are based on a monocular perspective, which requires training and testing the network with monocular data from multiple viewpoints, which increases

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the model's complexity and slows down the process of depth estimation [9]. A second type of endoscopic depth estimate (EDE) method is based on binocular perspective, which was originally proposed for natural scenes [10], while the effect is considerably diminished since the texture characteristics of endoscopic images are not as distinct as in natural scenes [11–13].

Although the EDE methods mentioned above had achieved some preliminary results, they are still quite challenging due to the scarcity of high-quality training datasets and processing time. First, actual endoscopic image datasets are challenging to create from technical and ethical aspects, such as patients' privacy, great cost, medical regulations, and the lack of ground truth annotations [14]. Secondly, actual endoscopic images are characterized by poor quality, low resolution, or quite noisy due to the large field of view, locally overexposed/underexposed, and irregular movement of the internal organs under examination. Lastly, EDE has stringent real-time requirements. Most published results are far from satisfying this need for practical applications [11,12,15]. A number of recent studies have proved that the performance of learning-based robot vision and control algorithms can be enhanced by generating simulation data, offering a novel insight into a variety of situations where real-world datasets are complicated and costly [16–19].

With regard to the paucity of binocular endoscopic datasets, we generate and render a high-resolution gastrointestinal model to generate a substantial amount of data for training the network and then test the trained network on actual data. In addition, we propose an end-to-end multi-scale binocular depth estimation (MMDENet) network by taking full account of the texture particularity of images of the endoscopic environment. We have performed ablation experiments to validate the efficacy of MMDENet, and the image data captured from a silicone bronchus Mannequin and actual endoscopic images are used to verify the network. We conclude by comparing it with state-of-the-art methods from error and run-time calculations, and an outlook is drawn. In summary, the contributions of this research are as follows:

- The creation of a high-accuracy 3D simulation model of the gastrointestinal environment provides an efficient tool for generating endoscopic image datasets with depth information. Based on real-world requirements, simulated gastrointestinal organs possess anatomically realistic diameters, flexion angles, polyps, and texture properties.
- In the feature extraction step, we propose a multi-scale feature extraction module, which assigns a particular weight to each feature map through the attention mechanism. This further obtains feature representation with rich semantic contextual information in order to enhance the network's matching accuracy in the ill-posed region.
- In the disparity refinement step, we propose a multi-dimensional information-guidance refinement module, which considers the fusion of left image photometric consistency bias, disparity map geometric consistency bias, and feature map characteristic consistency bias to refine the disparity map.

2. Related work

2.1. Depth estimation of natural scenes

Given the theoretical maturity and simplicity of the hardware, binocular stereo-matching methods are used for depth estimation tasks in many natural scenes. The deep learning-based stereo-matching algorithm has recently developed rapidly in two aspects, computing the matching cost and optimizing the disparity map.

In calculating the matching cost, Zbontar and Lecun's first proposal for computing the matching cost was to use the deep neural network features [20]. Wang et al. proposed the Anytime Stereo Network (AnyNet), which adopts variable resolution in order to speed up the speed and improve the accuracy [21]. The Pyramid Stereo Matching Network (PSMNet) optimized the cost by implementing a spatial pyramid pooling module of spatial pyramids [22]. The Group-wise Correlation Stereo Network (GwcNet) extended PSMNet by adopting the cost of group correlation cost and enhancing the 3D stacked hourglass lattice [23]. The Stereo Dilated Residual Network (StereoDRNet) introduced Vortex-Pooling architecture in the feature extraction process to increase the receptive field of cost calculation [24]. The Efficient Stereo Matching Network (ESNet) relied solely on 2D convolution and calculated the multi-scale cost volume efficiently using a warping-based method to improve the performance in regions with fine detail [25].

In the case of disparity map refinement, Shaked et al. resolved the mismatch situation by calculating the reflection confidence of the disparity map [26]. The Guided Aggregation Network (GANet) adopted semi-global clustering and locally-guided clustering layers for the purpose of optimizing delicate structures [8]. Liang et al. combined additional information, such as semantic features, to refine disparity maps [27]. Khamis et al. proposed a real-time stereo matching network (StereoNet), which recovers the details of disparity under the guidance of the original color images [7]. StereoDRNet fused the left image's photometric and geometric consistency bias to refine the disparity map [24]. Xu et al. proposed the Adaptive Aggregation Network (AANet), an in-scale cost aggregation method based on sparse points to replace complex 3D convolutions, which has a fast speed and high accuracy [28]. Wang et al. used recurrent units to iteratively update disparity estimations at high resolution to avoid the error accumulation problem encountered in the coarse-to-fine paradigm [29].

All these methods have achieved specific achievements in natural scenes. However, they all face the challenges mentioned above when applied to the endoscopic environment. In this work, we introduce an attention mechanism based on StereoDRNet, which obtains the weights of the feature maps to further improve the matching accuracy in ill-posed regions. Furthermore, we present a feature map characteristic consistency bias based on the photometric and geometric consistency to refine the disparity map in three dimensions.

2.2. Endoscopic depth estimation

Early EDE methods have been carried over from depth estimation methods that have been developed in natural scenes. These methods can be classified into monocular methods and binocular methods.

For monocular EDE methods, Mahmood et al. proposed a joint deep CNN and a conditional random field framework for monocular depth estimation [30]. Mahmood et al. used conditional generative adversarial networks (GANs) for EDE [31]. Anita et al. trained a pix2pix GAN to directly convert monocular endoscopic images into depth maps [32]. Oda et al. applied lambertian surface translation to an endoscopic image in order to remove texture and reflections on the surface of an organ. Then they estimated the depth using a fully-convolutional network (FCN) [33]. Liu et al. proposed a method based on multi-loss rebalancing for joint estimation of depth and motion from a sequence of monocular endoscopy images [34].

Regarding binocular EDE methods, Wang et al. proposed a Gaussian weighted AD-Census transform and an improved cross-adaptive region-based stereo-matching algorithm to achieve 3D measurement and reconstruction in a narrow endoscopic space [35]. In our previous work, we introduced a 23-layer deep CNN approach with a scale-invariant loss function to generate stereo depth maps in real-time [36]. Long et al. designed a transformer-based stereoscopic depth perception for efficient depth estimation and a lightweight tool segment to handle tool occlusion [37]. Zhang et al. introduced U-net fused by the Laplacian pyramid to improve the network's ability to extract intermediate features and used a convolutional block attention module to optimize network parameters for EDE [38]. Shi et al. proposed two self-supervisions, i.e., Adaptive Cross Supervision (ACS) and Adaptive Parallel Supervision (APS) in order to learn a convolutional neural

network with two branches for EDE [39]. To address the problem of imperfectly rectified stereo images, Luo et al. propose a vertical correction module and introduce a residual mask to improve the accuracy of the network output [40].

Most of these methods are adopted directly from depth estimation methods in natural scenes, and as a result, the reconstruction accuracy is typically low. The main reason for this is that the texture characteristics of endoscopic images are not salient enough compared to natural scenes. Considering the endoscopic image features in full, we propose a multi-scale feature fusion mechanism and adopt multi-dimensional information to the network module, which can maximize the use of information from endoscopic images in depth estimation.

2.3. Endoscopic datasets acquisition

Supervised deep learning is well known to require big data for the training of networks in order to achieve high precision. The paucity of labeled datasets in the endoscopic setting has become one of the biggest barriers to the development of high-precision learnable EDE methods.

Some research focuses on generating endoscopic simulation datasets with labeled depths for supervised learning. Typically, these datasets are generated from simulation modeling and rendering, Mannequin phantom or silicone, and animal organs (e.g., porcine colon). Mahmood et al. generated the training datasets from a digital colon with blender 3D modeling and a silicone colon phantom with X-ray computed tomography (CT) data. They then generated the testing datasets from a porcine colon via a CT scanner and a registered optical endoscopy for depths and 2D images, respectively [30]. To generate more realistic images, cinematic rendering technology was utilized to mimic the light propagation and interaction with objects during the 3D modeling [41]. In addition, they proposed an inverse flow framework, using GAN training to make the simulation and actual medical images indistinguishable from each other [14]. Jeong et al. used CT colonography to generate depth images from segmented models of the colon and acquired endoscopic images via a virtual endoscopy. Next, they then applied the Cycle-Consistent Generative Adversarial Networks (CycleGAN) to generate endoscopic simulation datasets based on CT data [42]. Incetan et al. presented a virtual endoscopy environment, that is capable of generating labeled image data at depth [43]. Ozyoruk et al. introduced a simultaneous endoscopic localization and mapping dataset with one Canon Aquilion Precision CT scanner and two commercial 3D scanners for both ground truth images and depths [44]. In some datasets, 3D endoscopic data were reconstructed from CT images, whereas depth information is obtained through fine adjustments performed manually using the Slicer software [45]. To ensure the complete reproducibility of experimentation and objective comparisons with other methods, Wijata et al. highlighted the critical significance of proper training/test dataset splits in order to address potential issues related to incorrect design [46].

Although CT and 3D scanners are capable of obtaining accurate depth information, they are expensive and laborious to acquire a substantial amount of actual endoscopic data and are not human-ready due to privacy and medical regulations. In addition, these simulation datasets are all developed for monocular EDE methods, whereas datasets for binocular methods are yet to be seen in the public reports. The goal of this work is to generate the datasets for binocular EDE methods in order to bridge the gap.

3. Method

3.1. Binocular datasets

The binocular stereo principle states that the depth of the object depends on the disparity between the two pairs of images and the camera parameters. The disparity is the horizontal shift from the left image to the right, which is commonly at the pixel level and is sometimes referred to as a disparity map. Depth is calculated as follows:

$$D = \frac{f * b}{d * p} \tag{1}$$

where, f is the focal length of the camera, b is the baseline, d is the disparity, D is the depth and p is the pixel size of the camera sensor. Because the camera parameters are fixed, once disparity can be inferred from the pair of images, depth is calculated accordingly. The goal of this work is to train the neural network to infer disparity.

In order to generate the binocular training datasets, we construct a 3D gastrointestinal model with high-accuracy rendering and then use a virtual binocular camera to emulate the image capture process. The proposed procedure can simultaneously generate a large amount of 2D image pairs with labeled depth information.

The trained network is evaluated on two types of test datasets, a Mannequin dataset as well as open source endoscopic datasets (Kvasir [47] and ETIS-LaribPolypDB [48]). The Mannequin dataset is captured from a silicone model of the gastrointestinal organs, via a prototype binocular endoscope to obtain 2D images and a displacement sensor to assess the true depth data. The Mannequin dataset is primarily used to quantitatively test the prediction results of the trained network. Open source datasets show qualitative results as they contain no ground truth depth information. Table 1 presents a comparison of the diverse endoscopy data employed in this study.

3.1.1. Training datasets

All training datasets are generated from 3D simulation, actual endoscopic images, and CT data. First, we use the open-source blender [49] software package to construct and render a high-precision 3D gastrointestinal model in Fig. 1(a). Simple grids and curves are used to generate the 3D model framework. Furthermore, in order to ensure that the 3D model has anatomically realistic texture properties, we extract texture maps from different features (e.g., veins and tissues) from the actual endoscopic images in the open-source dataset Kvasir and then project them on the 3D model in a uniform and flat pattern. Lastly, given the lack of high spatial frequency information in gastrointestinal simulation models, in order to incorporate these features [43], we adopt the spatial texture reconstructed from the CT data, as can be seen in Fig. 1(c).

Images in the training datasets are then generated via a virtual imaging process, as shown in Fig. 1(b). It is easy to adjust the virtual endoscope to mimic different endoscopes. In this work, we set the same parameters as an as-built prototype binocular endoscope in our laboratory to test the trained system. We use a Bessel polynomial to adapt to a camera's path constraints, and an illumination source is added to the device to control illumination. This allows the camera to be set up as a multi-view mode to generate the binocular view rendering while changing the texture maps and camera angles to enrich the datasets. For training and testing the network, we generate 3000 image pairs. 2550 stereo pairs are used for training and an additional 450 pairs are used for testing. The dataset split is provided directly by Ye et al. and performed through a random selection of the images [50]. The rendered images have a resolution of 800 × 800. Example images from the datasets can be found in Fig. 1(d).

3.1.2. Testing datasets

To test the trained network's accuracy in the application of an electronic binocular endoscope, we generate a Mannequin dataset on a model of a silicone gastrointestinal organ (KAY-LV48) in this paper. In order to capture 2D images, we use an as-built endoscope (see Fig. 2). A prototype of the binocular endoscope is constructed in our laboratory with two tiny cameras (left and right, baseline 2 mm, focal length 1.5 mm, sensor pixel size 0.002 mm) and an illumination source in the distal tip of the endoscope. A linear displacement sensor (YKEK KPZ 50 mm) is attached to the endoscope for measuring the depth information simultaneously. The camera is capable of capturing images/videos

Table 1
Detailed information of different endoscopy datasets used in this study.

Dataset	Virtual/real endoscopy	Ground truth depth available	Training	Testing	Data size
Synthetic	Virtual	Yes	Yes	Yes	3000
Mannequin	Real	Yes	No	Quantitative	N/A
Kvasir [47]	Real	No	No	Qualitative	2000
ETIS-LaribPolypDB [48]	Real	No	No	Qualitative	192

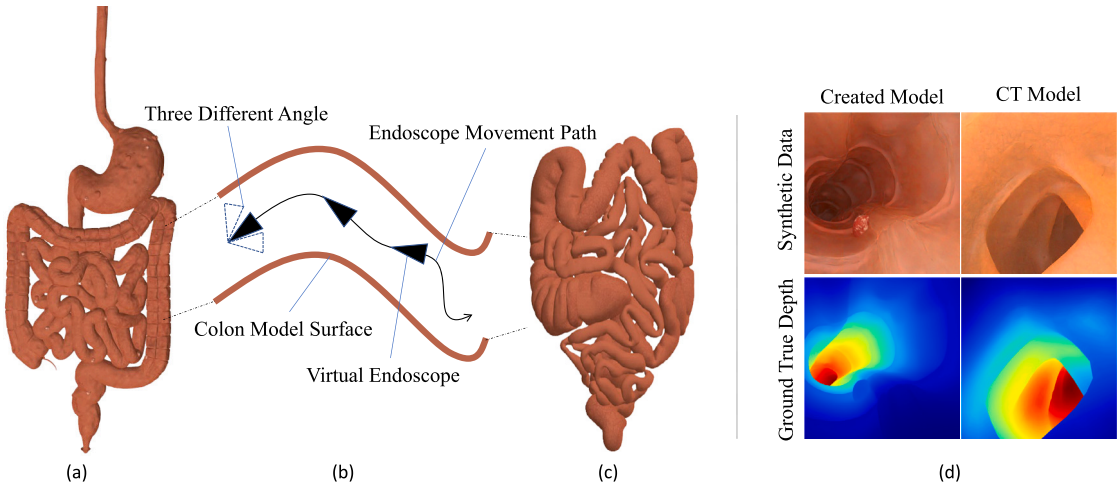

Fig. 1. The training datasets consists of simulation images from a 3D model and CT data, (a) the 3D gastrointestinal simulation model, (b) data acquired in a virtual endoscope with different angles and motion paths, (c) a reconstructed model using CT data [43], and (d) exemplary data with image and depth information.

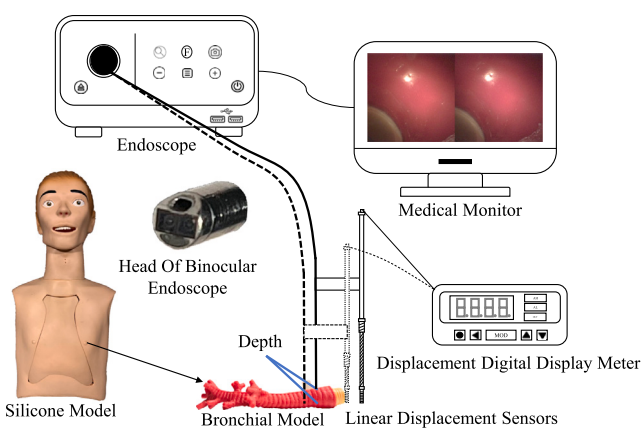

Fig. 2. The generation of silicone Mannequin datasets. An as-built binocular endoscope captures images, and the depth data is acquired via a displacement sensor.

from two synchronized cameras using the same exposure time (30 fps). To measure depth information simultaneously, a linear displacement sensor (YKEK KPZ 50 mm) is attached to the endoscope. Using the displacement sensor, the measured point depth difference (dashed line) can be given by the reference point (solid line). By referencing the same point, we can obtain depth information from all captured 2D images. We acquired four pairs of images with a resolution of 800 × 800, and the corresponding ground-truth depth values of 12.4 mm, 17.97 mm, 19.60 mm, and 22.57 mm respectively.

To further test our model, we use several images from the Kvasir dataset containing images of the inner gastrointestinal tract [47] as well as the ETIS-LaribPolypDB dataset for the detection of polyps [48]. From the Kvasir dataset, we select 4 images of normal and ulcerated tissue and 2 images containing polyps from the ETIS-LaribPolypDB dataset. All of the open source datasets have only monocular images, so we manually crop the selected images to generate the image pairs by allowing the maximum disparity to be 20 pixels, following the distribution of statistical disparity in training datasets and preserving the same image resolution 800 × 800.

3.2. MMDENet

The binocular stereo EDE can be simply split into two stages: image matching and disparity estimation. One of the most widely used image-matching methods is matching based on image feature information. The proposed MMDENet framework can be seen in Fig. 3(a). First, our work uses a multi-scale module to extract input features from stereo image pairs in order to compute the matching cost and form the cost volume. The model then uses a 3D convolutional neural network and the disparity regression function Softargmin has generated the preliminary disparity map of interest. Lastly, the multidimensional module refines the preliminary disparity map through multidimensional data.

3.2.1. Multi-scale feature extraction

In this paper, we design a multi-scale feature fusion module based on the similarity of left and right binocular images. First, as shown in Fig. 3(b), three convolutions layers of 5 × 5 are used to downsample the high-resolution input image to obtain a 32-channel low-resolution feature map of the image. The initial feature map then extracts the deep feature representation across six clusters from a standard residual block, where each residual block consists of a 2-dimensional convolution, a batch normalization [51], and a rectifying linear unit [52]. Vortex Pooling [53] is chosen from the learned local feature maps in order to encode more contextual information, with pooling parameters as global, 3, 5, 7, followed by 1 × 1 convolution and 3 × 3 convolution with the corresponding dilation factors, respectively. To obtain a group of feature maps, we concatenate the feature maps obtained from the four scales.

As the amount of information contained in each scaled feature map is different, SENet learns the weights of the feature map across the network in order to get better results [54]. Inspired by it, this work assigns different weights to each scale feature map. Global mean

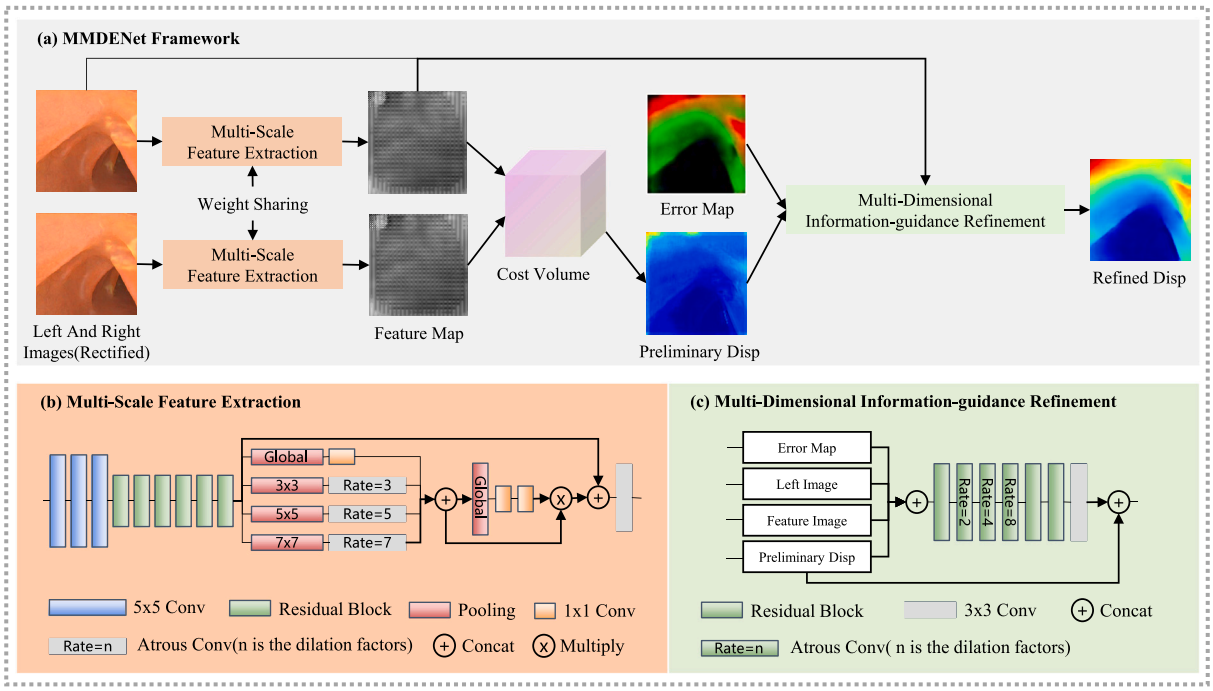

Fig. 3. Overview of the proposed MMDENet framework. Include a feature extraction module integrating multi-scale contextual information and a multi-dimensional information-guided disparity refinement module.

pooling transforms the group of feature maps into one-dimensional feature vectors. In the first convolution layer 1×1, the number of channels is compressed, and the second convolution layer 1×1 restores the number of channels after the ReLU function is turned on. The weighted feature map group is obtained by multiplying the weight coefficient with the corresponding feature map. The initial feature map is then concatenated with the group of weighted feature maps via a skip connection. To obtain the final merged feature map, we compress the number of channels to 32 via a convolutional layer of 3×3.

3.2.2. Matching cost and disparity calculation

In the previous step, a merged feature map of multi-scale contextual information was obtained for each pixel. To compute the matching cost, we subtract the left-hand feature map and the right-hand feature map at all possible disparities and construct it into a 4D raw cost volume. This operation is expressed mathematically as follows:

$$F(d,h,w) = L(h,w) - R(h,w+d) \tag{2}$$

where L is the left feature, R is the right feature, h and w are the height and width of the feature map, and d is the distance between the two translations.

In order to aggregate the raw cost volume, this work selects four 3D convolutions with a filter size of $3 \times 3 \times 3$, batch normalization, and leaky activation of ReLu. The final 3D convolutional layer does not use batch normalization or activation. The three-dimensional convolution can aggregate the raw cost volume from three dimensions of disparity, height, and width based on keeping the full information [21]. The regularized cost volume is ultimately generated via the above operations.

In traditional disparity computation, a Winner-Takes-All (WTA) strategy is used by choosing the smallest value in the disparity dimension of the cost volume as the optimum disparity value of the current pixel [55]. Compared to the WTA strategy, Softargmin's function considers all possible disparities instead of a single pixel. As such, improved smoothing and sub-pixel-level accuracy can be achieved. On the other hand, the differentiability of the Softargmin function guarantees that the network can be trained using the backpropagation mechanism [7].

The disparity of a pixel is equal to the weighted sum of the possible disparity values of the dot, and the weight is the probability of the disparity being possible. The mathematical expression is

$$d'(h,w) = \sum_{d=D_{\min}}^{D_{\max}} d \times \sigma\left(C_i(c,d,h,w)\right) \tag{3}$$

where d' is disparity map, C_i is a 4-dimensional feature tensor, $D_{\max}$ and $D_{\min}$ are the maximum disparity value and the minimum disparity value, c is the number of channels, d is the possible disparity value of each pixel in the image, σ is the Softmax function, and $\times$ is the dot product operation. We have now completed the preliminary single-channel disparity estimation.

3.2.3. Multi-dimensional information-guided refinement

Using Eq. (1) to get the disparity map has an unsatisfactory prediction result in the edge area of the object. The goal of disparity refinement is to find a meaningful representation of the image features to guide the subsequent disparity for precise matching. At this stage, the network uses some metric as a guide to inflate or erode disparity values to merge high-frequency detail.

To refine the disparity map in three dimensions, we present a feature coherence bias in the feature map based on photometric and geometric coherence. As can be seen in Fig. 3(c), we first warp the right image I_r to the left image I_l via the warp W and calculate the image reconstruction error map E for the left image I_l as:

$$E = \left| I_l - W\left(I_r, D_r\right) \right| \tag{4}$$

Then the raw disparity map, the left image, the left feature maps, and the reconstructed error map are concatenated.

The concatenated tensor is also passed through 6 residual blocks, in which the atrous convolution operation is used in each residual block in order to expand the filled sensing [56], and the dilation factors are set to be equal to (1, 2, 4, 8, 1, 1). Next, the tensor is passed through a 2D convolutional layer 3×3 and the preliminary disparity is concatenated to achieve an enhanced disparity effect.

3.2.4. Loss function

Due to the unique properties of the simulated datasets such as smoothness and limited texture, the prediction of scene depth on a global scale is fuzzy. We use scale-invariant error to measure the relation between local points in the scene, regardless of the L2 loss on the absolute global scale. The scale-invariant error function is defined by [36,57]

$$L\left(y, y^*\right) = \frac{1}{n}\sum_{i=1}^{n}\left(\log y_i - \log y_i^* + \alpha\left(y, y^*\right)\right)^2 \tag{5}$$

$$\alpha\left(y, y^*\right) = \frac{1}{n}\sum_{i}\left(\log y_i^* - \log y_i\right) \tag{6}$$

where y is the predicted disparity, y^* is the real disparity, $\alpha(y, y^*)$ is the average value of (y, y^*) interpolation in logarithmic space.

Considering the model consists of two stages, the total training loss is calculated by

$$L_{total}\left(y, y^*\right) = w_1 * L_1\left(y, y^*\right) + w_2 * L_2\left(y, y^*\right) \tag{7}$$

$$L_i\left(y, y^*\right) = \frac{1}{n}\sum_{i} d_i^2 - \frac{\lambda}{n^2}\left(\sum_{i} d_i\right)^2 \tag{8}$$

$$d_i = \log y_i - \log y_i^* \tag{9}$$

where w_1 and w_2 are the weights of the two stages respectively, we set them as 0.6 and 0.4. d_i is the disparity value on pixel i. λ is a weighting factor, used to balance the absolute and relative loss. If $\lambda = 1$, the loss is a proportional-invariant relative error; else if $\lambda = 0$, the absolute L2 loss is used. We set $\lambda = 0.5$ to consider both absolute and relative information, and to make the image more realistic.

4. Experiment

In this section, we first introduce the training details of the MMDENet. The ablation experiments for the proposed two modules are then performed. Our model is evaluated both quantitatively and qualitatively using Mannequin captured data with silicone bronchus and actual endoscopic images. We conclude with a comparison to some state-of-the-art methods.

4.1. Training settings

The parameters of the trained network will be affected by the disparity distribution of the dataset, so we statistically analyze the datasets used. As can be seen in Fig. 4, the disparity distributions of the training and test datasets are uniform. Most of the disparity is within [25, 200] pixels, and the disparity of 75 pixels has the highest probability, while other values occupy less than 5%. According to Eq. (2), the corresponding depth distribution is in the range [7.81, 62.5] mm, of which the depth of 20 mm is equal to a disparity of 75 pixels.

For the deep learning model, we use the PyTorch [58] platform to perform the training and testing on a single Nvidia RTX 3060Ti GPU. In all of our experiments, we optimize using RMSprop with an exponentially decaying learning rate initially set to 1e−3. The training process runs for 20 epochs in total. Each epoch consists of the initial disparity estimate and disparity refinement step. The maximum and minimum disparity values are set to 192 and 24, respectively, based on the statistical results from the datasets. The total training time for Dataset was 3.2 h, and all results are checked on 100 random datasets on average.

There are six metrics that are used to evaluate the performance of the model, Endpoint error (EPE), K pixel error (KPE), Absolute and Relative Error (Abs Rel), Square Relative Error(Sq Rel), Root Mean Square Error(RMSE), and Root Mean Squared Log Error(RMSE Log), as can be seen in 2.

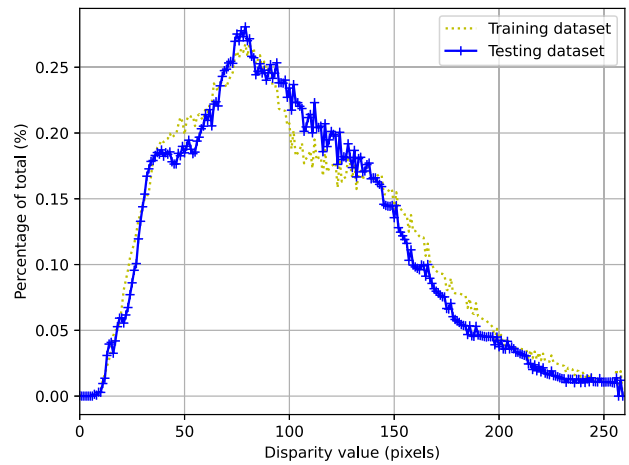

Fig. 4. Disparity distribution of the training datasets and the testing datasets.

Table 2
The error and accuracy metrics for disparity evaluation. d and d^* respectively denote the predicted disparity value and the corresponding ground truth value, k represents a pixel value, **D** represents a set of predicted disparity values.

Metric	Definition
EPE	$\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}\lvert d^* - d\rvert$
KPE	$\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}(\lvert d^* - d\rvert > k)$
Abs Rel	$\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}\lvert d^* - d\rvert / d^*$
Sq Rel	$\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}\lvert d^* - d\rvert^2 / d^*$
RMSE	$\sqrt{\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}\lvert d^* - d\rvert^2}$
RMSE Log	$\sqrt{\frac{1}{\lvert\mathbf{D}\rvert}\sum_{d\in\mathbf{D}}\lvert\log d^* - \log d\rvert^2}$

Table 3
Ablation experiment to validate the multi-scale feature fusion module and the multi-dimensional information guidance module. The baseline refers to the stereo-matching calculation module [36].

Method	Network settings		EPE (pixel)	RunTime (s)
	Feature extraction	Refinement		
1	Baseline	Baseline	29.28	**0.0271**
2	Multi-scale	Baseline	28.20	0.0307
3	Baseline	Multi-dimensional	27.57	0.0315
4	Multi-scale	Multi-dimensional	**26.74**	0.0338

4.2. Ablation study

To check the functionality of the proposed modules we carry out the following ablation experiments as shown in Table 3. We construct four methods of network models, and each method has only a single variable functional module relative to the others. Method 2, for example, is used to verify the multi-scale feature extraction module, when compared with the results of method 1. The first method serves as the benchmark results without the use of the proposed modules. The metrics that we used to compare each method are the EPE and execution time for a single frame.

Experimental results show that the EPEs of method 2 with the multi-scale feature fusion module and method 3 with the multidimensional information guidance module are 28.20 and 27.57, respectively, lower than the baseline value of 29.28. Thus, adding the two proposed modules may reduce the matching error. There is a slight increase in runtime compared to the baseline as both refinement modules increase the complexity of the network parameters. We replace the feature extraction and disparity refinement modules in the MMDENet framework (method 4) with the proposed ones, and the EPE is decreased by 2.54 (about 8.7%) relative to the baseline value. A small step of 0.0067 s in

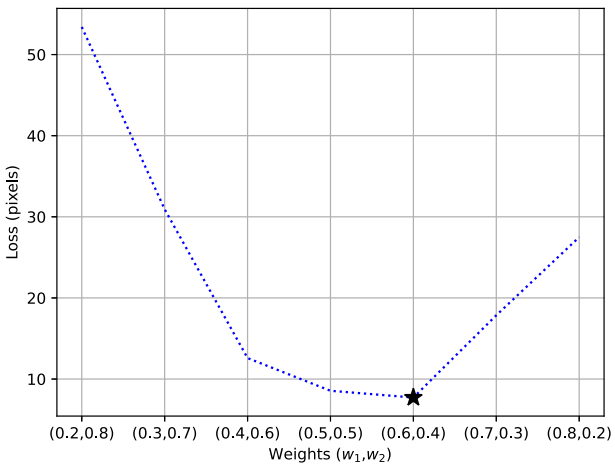

Fig. 5. Losses for different weight combinations of w_1, w_2. The one with an asterisk mark is selected for the subsequent experiments.

Table 4
Qualitative evaluation of the MMDENet in terms of MAE and MRE. GT is the ground-truth depth value, MAE is the average absolute error, and MRE is the average relative error.

Group	Average depths (mm)			GT (mm)	MAE (mm)	MRE (%)
	1 pixel	3 pixels	5 pixels			
1	13.32	13.32	13.32	12.40	0.92	7.41%
2	19.43	19.44	19.47	17.97	1.48	8.22%
3	20.39	20.39	20.39	19.60	0.79	4.03%
4	22.55	22.55	22.57	21.12	1.33	6.28%
Avg	–	–	–	–	**1.09**	**6.27%**

the execution time still allows the image processing speed to reach 30 fps and satisfy the real-time requirements.

The loss function hyperparameters, denoted as w_1 and w_2, represent the weights assigned to the disparity estimation stage and the disparity refinement stage of the network, respectively. In order to demonstrate how these hyperparameters were determined, we conducted an ablation study with different combinations of these two hyperparameters, and observed the corresponding training results, as depicted in Fig. 5. Note that the network achieved the lowest loss when the hyperparameters were set to $w_1 = 0.6$ and $w_2 = 0.4$. This suggests that the disparity estimation stage holds slightly more significance than the subsequent disparity refinement stage. Additionally, it is observed that larger differences in weights between these two stages result in reduced performance.

4.3. Qualitative and quantitative evaluation

In order to assess the accuracy of the depth estimation, we tested the trained MMDENet model in predicting the depths of pairs of endoscopic images in the silicone bronchus dataset as well as real endoscopic images from open source datasets (Kvasir, ETIS-LaribPolypDB).

Table 4 shows the results of the quantitative analysis for testing image pairs in the dataset of silicone bronchi. The predicted depth result takes the mean value of different small areas, for example, in the range of 1×1 pixels, 3×3 pixels, and 5×5 pixels, respectively. The statistical results show that the MAE of our depth prediction is 1.09 mm and that the MRE is approximately 6.27%. The greatest accuracy is obtained when the depth of the GT is 19.6 mm, where the MAE is 0.79 mm and the MRE is 4.03%. It should be noted that most of the depths in the dataset are around 20 mm, implying that a larger dataset may be useful in increasing prediction accuracy.

In addition, we tested the MMDENet on real endoscopic images from the Kvasir and ETIS-LaribPolypDB endoscopic datasets [47,48]. As there is no ground truth depth available for these images, the results are only qualitative as shown in Fig. 6.

This qualitative result indicates that the proposed MMDENet can still generate coarse depth maps and has good performance in cross-dataset validation, even though it was purely trained on the simulation dataset.

4.4. Comparative experiments

In order to compare our model with state-of-the-art depth estimation methods, we implemented the same depth prediction tasks using the simulation dataset, in the methods StereoNet [7], AnyNet [21], AANet [28], CFNet [59], and Lac-GwcNet [60] respectively.

The experimental results shown in Fig. 7 show that the results generated by the AnyNet and AANet methods are far from GT, note that the pseudo color display results of the StereoNet and Lac-GwcNet methods do not match GT, the CFNet approach produced similar results to GT but somewhat blurred, compared to the structures of the counterpart networks, our proposed MMDENet is able to predict better in detail, e.g., edges in the disparity map. The proposed multi-scale and multidimensional modules appear to effectively tune the depth disparity map in more versatile situations since the pseudo color visualization results of our predicted depth maps are in general closer to the GT depth maps of the images.

Table 5 further demonstrates the comparison of the different models on seven metrics. Statistical results show that the first six metrics are the smallest of all the methods, which further indicates that our model achieves the highest accuracy with respect to the disparity map. The error in the EPE is reduced by 59.66% compared to the CFNet model, while the execution time per frame is reduced from 0.0399 s to 0.0338 s. Of all networks, the velocity of our model and the AnyNet satisfy 30 fps.

To statistically investigate the performance differences between the proposed method and other algorithms, we conducted a Friedman test [61] on the average EPE metric. The test is conducted by dividing the testing dataset into five groups randomly, each containing ninety samples. After running each network, we calculated the average EPE value for each group. These EPE values are imported to the Friedman test in R, which outputs a p-value to indicate if all algorithms perform equally. Typically, if the p-value is larger than 0.05, the test attempts are more equal. Our experiment shows a p-value of 0.0014, implying a significant difference in performance compared with the other algorithms.

5. Discussion

In this work, we propose a multi-scale, multi-dimensional deep-learning network for binocular endoscopic depth estimation. This may improve the quality of the output disparity map without a reduction in running speed. The goal of a feature extraction network incorporating multi-scale contextual information is to improve the correspondence accuracy of ill-posed regions. Three-dimensional information, including the left image photometric consistency bias, the disparity map geometric consistency bias, and the feature map characteristic consistency bias, is considered to guide the refinement of a rough initial disparity map.

To cope with the paucity of binocular endoscopic datasets for efficient training, we generate 3000 image pairs from 3D simulation and modeling, real endoscopic images, reconstructed CT data, and texture maps. The ablation experiments are performed, and the statistical results show that the endpoint error of the prediction results is reduced by approximately 8.7% whereas the time cost per frame is approximately 0.033 s. In this paper, we evaluate our model using two methods, (1) capturing endoscopic images of a silicone bronchus through an in-house binocular endoscope, and (2) using real-world images from open-source medical endoscopic datasets. The qualitative

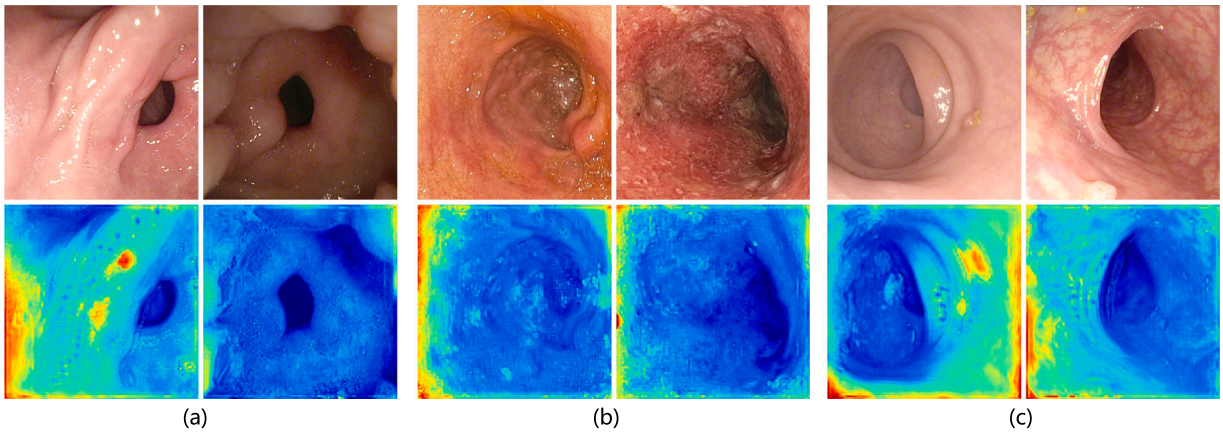

Fig. 6. Qualitative depth estimation results of the trained model on actual endoscopic datasets, (a) normal images and (b) ulcer images [47], (c) polyps images [48].

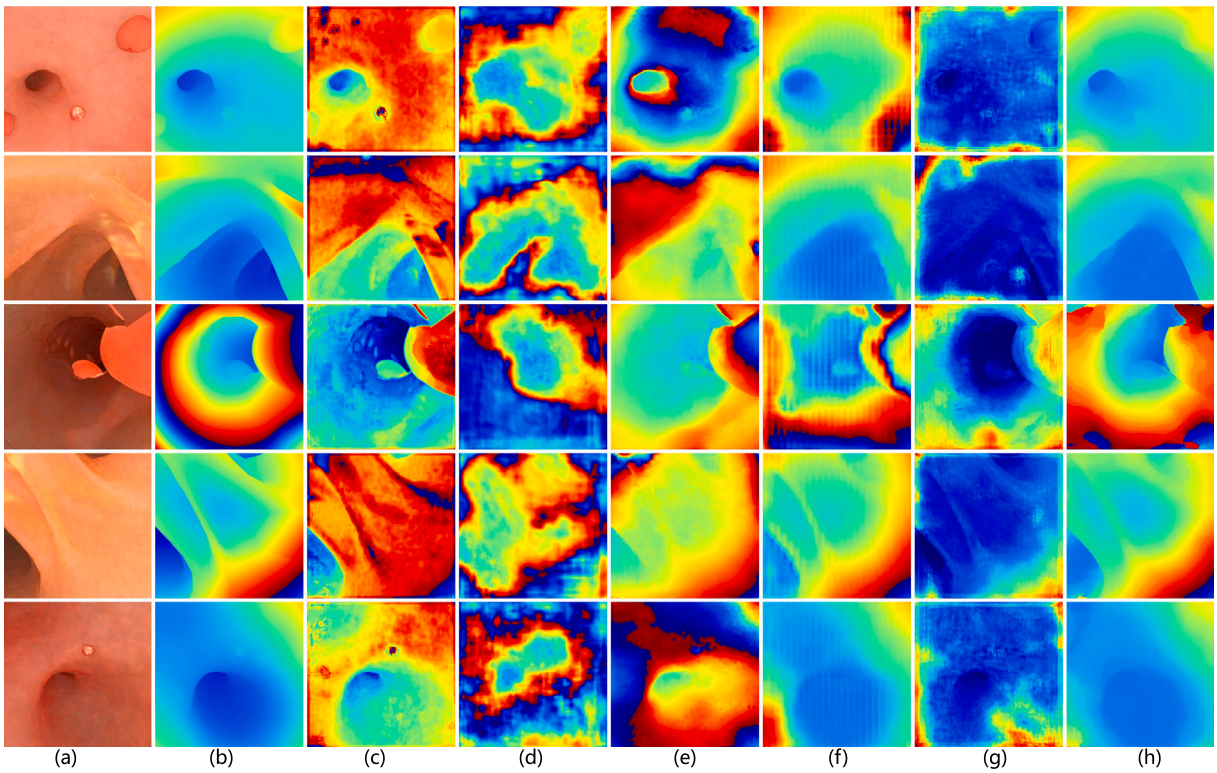

Fig. 7. Comparative experimental results on the simulation dataset, (a) input images and (b) the ground-truth depth maps. The predicted depth map results on (c) StereoNet [7], (d) AnyNet [21], (e) AANet [28], (f) CFNet [59], (g) Lac-GwcNet [60], (h) our proposed MMDENet.

Table 5
Depth prediction performance comparison with state-of-the-art methods in terms of six metrics. The top two results are highlighted in bold and underlined.

Method	KPE (%)			EPE (pixel)	Abs Rel	Sq Rel	RMSE	RMSE Log	RunTime (s)
	1 pixel	3 pixels	5 pixels						
StereoNet	0.932	0.844	0.794	27.611	0.406	16.761	30.751	0.422	0.039
AnyNet	0.977	0.932	0.887	30.719	0.532	27.024	35.222	0.434	**0.019**
AANet	0.885	0.767	0.695	27.685	0.517	35.521	31.916	0.389	0.105
CFNet	0.823	0.553	0.411	13.468	0.209	12.494	18.081	0.246	0.052
Lac-GwcNet	0.995	0.986	0.876	21.163	0.391	11.594	29.632	0.322	0.174
MMDENet(ours)	**0.591**	**0.297**	**0.204**	**5.433**	**0.049**	**1.627**	**9.612**	**0.135**	0.034

results show that the depth information agrees with the true image, and an accuracy rate of 93.38%, demonstrating that the generated low-cost endoscopic datasets can be used for network training with high performance. In this section, we compare the performance of our proposed MMDENet with state-of-the-art network models in terms of six metrics, by demonstrating the trained MMDENet, the highest accuracy can be obtained while maintaining a real-time processing demand.

There are limitations of our method that need to be taken into consideration in future work. First, our virtual imaging results in Fig. 1(d) show slight unreal, which is caused by the distorted information due to the methods or equipment used in the modeling or 3D scanning process is not advanced. In future work, we will attempt to explore a more advanced rendering engine (Unreal Engine 5) for the generation of virtual data. Second, our model uses a single Nvidia RTX 3060Ti GPU for its execution, which is difficult to incorporate into an endoscope host system. Future work will attempt to prune the model or consider an effective lightweight network model in order to achieve industrialization.

CRediT authorship contribution statement

Xiongzhi Wang: Conceptualization, Methodology, Software. **Min Wei:** Data curation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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